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# Abdelhak Bourdim

## Senior Embedded Software Engineer — C/C++ • RTOS • Embedded Linux

**Date of Birth:** August 20, 1969      **Nationality:** French      **Status:** Married

**Driving License:** B • Eligible for Luxembourg cross-border permit      **Email:** abdelhak.bourdim@gmail.com

**Phone:** +33 6 19 51 51 73      **Location:** Paris region, France

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**Mobility:** Luxembourg (cross-border from France) — Available immediately

## PROFILE

Embedded software specialist with **over 20 years of experience** developing firmware, middleware and application layers — bare metal, RTOS and Embedded Linux — for safety-critical and industrial systems. Design of low-level drivers, communication stacks (CAN, BLE, USB, TCP/IP) and embedded applications on ARM Cortex, STM32, Microchip and NXP architectures.

Certified solutions for **aerospace** (Safran, Thales — DO-178), **medical devices** (GE Healthcare, Sorin — IEC 62304), **automotive** (Valeo, Arkamys), **IoT cybersecurity** (Enedis — Linky), **payment systems** (Cartadis) and **smart EV charging V2G** (E2-CAD — ISO 15118). Trilingual French/English/Arabic, a strong asset for Luxembourg's multilingual market.

## PROFESSIONAL EXPERIENCE

### E2-CAD — Eragny, France

May 2025 – Mar. 2026

#### Embedded Software Engineer — EV Charging / V2G

*EV Charging Station — Iotecha IP*

- Reverse engineering and source code analysis (static and dynamic architecture)
- Embedded code review on 3 boards (Microchip PIC, STM32), debug tools (ST-Link, nsprog)
- Communication protocol analysis between Linux modules and embedded boards (RS232, I2C, Saleae)
- Python extension development and real-time HTML web report generation
- Test bench design and setup: Chargebyte tools integration, EV simulator
- Debug and commissioning of charging stations (PLC, HomePlug Green PHY, ISO 15118, OCPP)

**Env:** Embedded Linux, C/C++, Microchip PIC, STM32, PLC, HomePlug Green PHY, ISO 15118, OCPP, CAN, TCP/IP, Saleae, HackRF One, Python, Bash

### Safran — Creteil, France

Mar 2024 – May 2025

#### Embedded Software Engineer — Aerospace (SSPC)

*Solid State Power Controller — Low-level Drivers*

- Development and update of low-level C drivers for the SSPC system
- Logging tools setup for diagnostics and exchange traceability
- Writing and executing on-target validation tests with Lauterbach TRACE32
- Bash/Awk scripting for test process automation

**Env:** C, Lauterbach TRACE32, Bash, Awk

### Asterios Technologies — Massy, France

Aug 2023 – Feb 2024

#### Embedded Software Engineer — Aerospace (SOM Tests)

- C validation test development for SOM module (T1042 processor)
- Processor errata analysis and software workaround identification
- Test documentation writing per project requirements

**Env:** C, Bash, Python, Lauterbach TRACE32, Polarion

### Thales — Gennevilliers, France

May 2022 – Jul 2023

#### Embedded Software Engineer — Defense Radio

*LPM + N-MMR Projects — Mission Preparation & Multi Mode Receiver*

- Application layer development under Embedded Linux in C/C++ for 2 projects
- Configuration management and automation scripts (Bash, Curl, Tshark)
- HLT (High Level Tests) development and execution
- Bug fixing and network protocol analysis (Websockets)

**Env:** Embedded Linux, C/C++, Bash, Websockets, Curl, Tshark, Python

### Valeo — Creteil, France / Egypt

Mar – Apr 2022

#### Embedded Software Engineer — Test Audit

- Audit of test processes and tools (NI Mastre, Vector CANoe)
- Automation proposals and client/system requirements review

**Env:** NI Mastre, Vector CANoe

## Arkamys — Levallois-Perret, France

### Embedded Software Engineer — Automotive Audio

- GStreamer plugin and CAN driver development (ELM327), diagnostic tools
- Unit & integration testing (Valgrind), Buildroot build system

**Env:** Embedded Linux, C/C++, Buildroot, GStreamer, Valgrind, CAN, Python, Jira

## PROFESSIONAL EXPERIENCE (CONTINUED)

### Thales — Velizy-Villacoublay, France

Nov 2020 – Jun 2021

#### Embedded Software Engineer — Autonomous Train (ATO)

- 32-bit to 64-bit application migration on Embedded Linux platform
- IPC communication protocol definition and implementation
- Network diagnostic tools and integration test development

Env: Embedded Linux, C, UDP, TCP, Wireshark, Tshark, Python, Jira

### PK Vitality — Paris, France

Jun – Sep 2020

#### Embedded Software Engineer — Medical Device (CGM)

- HAL development (AFE AD5940, PMIC DA9070), QSPI driver, BLE NRF52840
- STM32 touchscreen TouchGFX, FreeRTOS, low power mode
- In vitro testing and Doxygen documentation (IEC 62304 compliant)

Env: STM32L4S9, NRF52840, C, C++, FreeRTOS, BLE, TouchGFX, Doxygen

### Zodiac Aerospace (Safran) — Montreuil / Vancouver, Canada

Feb 2017 – Mar 2020

#### Embedded Software Engineer — Aerospace (SSPC A350)

- Low-level layer adaptation: CAN, RS485 drivers, thermal memory (dsPIC33)
- uAFDX to CAN gateway implementation and diagnostic functions
- SSPC A350 application layer development and optimization
- Validation testing and aerospace-standard documentation
- Flight test campaign at UBC Vancouver, Canada (DO-178 certification)

Env: Microchip dsPIC33, MPLAB, TI Concerto, CCS Studio, C

### Enedis — Nanterre, France

May 2015 – Sep 2016

#### Embedded Cybersecurity Engineer — Smart Grid

- Security diagnostics and audit of Linky smart meter concentrators
- Penetration testing on Embedded Linux systems (Kali Linux, Nmap)
- Protocol sniffing and fuzzing (Wireshark, Tshark, Scapy, SDR)
- Code audit and review with manufacturers (Sonar, PC Lint)

Env: Kali Linux, Embedded Linux, Nmap, Wireshark, Scapy, SDR, Python, Bash

### Sorin — Clamart, France

Nov 2014 – Apr 2015

#### Embedded Software Engineer — Medical (BLE Pacemaker)

- Bluetooth Smart module validation, Serial Port Service Profile
- Smartphone communication protocol and Android application

Env: Cortex M0, Keil, C, BLE, Android

### General Electric Healthcare — Buc / USA

2012 – 2014

#### Embedded Software Engineer — Medical Imaging

- CANOpen stack integration (slave/master), CAN drivers and services
- CMX-RTX kernel porting, US team technical support

Env: TI TMS320, ARM Cortex A8, VxWorks, C, C++

### Photomaton / BCM / Vertical M2M / PGS — Paris

2010 – 2012

Payment systems and remote metering — Full embedded + PC development on NXP LPC (USB, RFID, GSM/GPRS, Bluetooth)

### Cartadis — Fontenay-sous-Bois, France

Oct 2001 – Dec 2009

8 years — Access control and payment terminals: drivers, embedded applications, USB/TCP/IP stacks, DES/RC4 encryption (NEC V853, Hitachi H8S, NXP LPC, ATMEL AT91)

## TECHNICAL SKILLS

|                    |                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------|
| Languages          | C, C++, Python, Bash, Assembly, Perl, Awk                                                   |
| OS / RTOS          | Embedded Linux, VxWorks, FreeRTOS, UCOSII, Tnkernel                                         |
| MCUs               | ARM Cortex (M0, M4, A8), STM32, NRF52, ESP32, Microchip PIC/dsPIC, NXP LPC, TI DSP/Concerto |
| Protocols / Bus    | CAN, CANOpen, SPI, I2C, RS232, RS485, UART, USB, BLE, AFDX, Ethernet, TCP/UDP               |
| Stacks             | TCP/IP, USB Host/Device/HID, BLE, CANOpen, RFID, GStreamer                                  |
| Security / Payment | Smart Card, RFID Mifare, DES, RC4, USB CCID, embedded pentesting                            |
| Tools              | Lauterbach TRACE32, IAR, Keil, MPLAB, CCS Studio, Green Hills, Buildroot, CubeMx, TouchGFX  |
| Management         | Git, Synergy, Jira, Polarion, Doxygen, Wireshark, Saleae                                    |

## STANDARDS & METHODS

## EDUCATION

**AFPA, Champs-sur-Marne, France**

Engineer — Electronic and Computer Systems Design

**Tenon Hospital, Paris, France**

Postgraduate — Biomedical Instrumentation

**University of Boumerdes, Algeria**

Engineer — Electronics, Communication specialization

## LANGUAGES

|         |                        |
|---------|------------------------|
| French  | <i>Native / Fluent</i> |
| English | <i>Professional</i>    |
| Arabic  | <i>Native / Fluent</i> |

## INDUSTRY SECTORS

Aerospace (Safran, Thales, Zodiac)  
Medical Devices (GE Healthcare, Sorin, PK Vitality)  
Automotive (Valeo, Arkamys)  
Energy / Smart Grid (Enedis — Linky)  
EV Charging / V2G (E2-CAD)  
IoT / M2M (Vertical M2M)  
Payment (Cartadis, Photomaton, BCM)

## VOLUNTEER WORK

**Workshop-DIY**

Founder — Chelles, France — since 2020

Robotics & STEM workshops for youth (ages 6-12+)  
ESP32 robot kit design (WiFi, BLE, servos, sensors)  
18+ workshops: Arduino, micro:bit, AI, 3D printing  
[workshop-diy.org](http://workshop-diy.org)